PICO: Ordering-Centric Cost-Based Optimization for ML Input Pipelines

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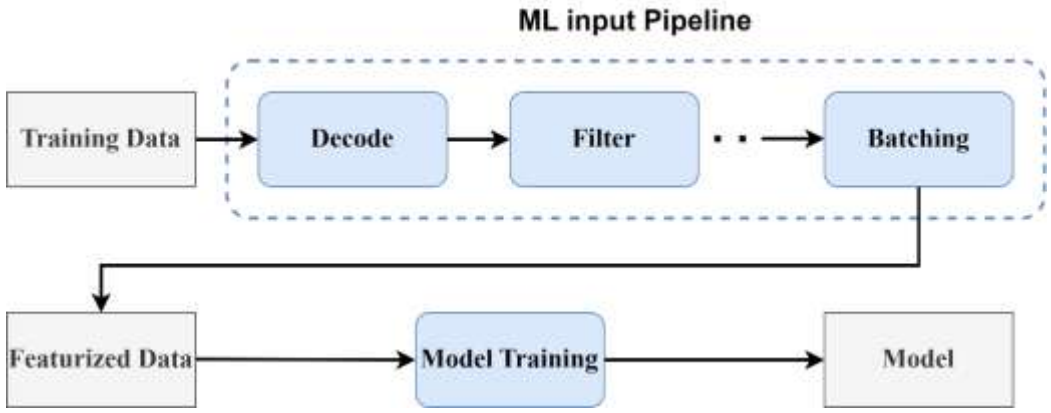
Outline

1. Background
2. Motivation
3. Optimal Reordering
4. Joint optimization
5. Experiment

ML input pipeline

An ML input pipeline performs the online, last-mile processing between stored data and the training loop.

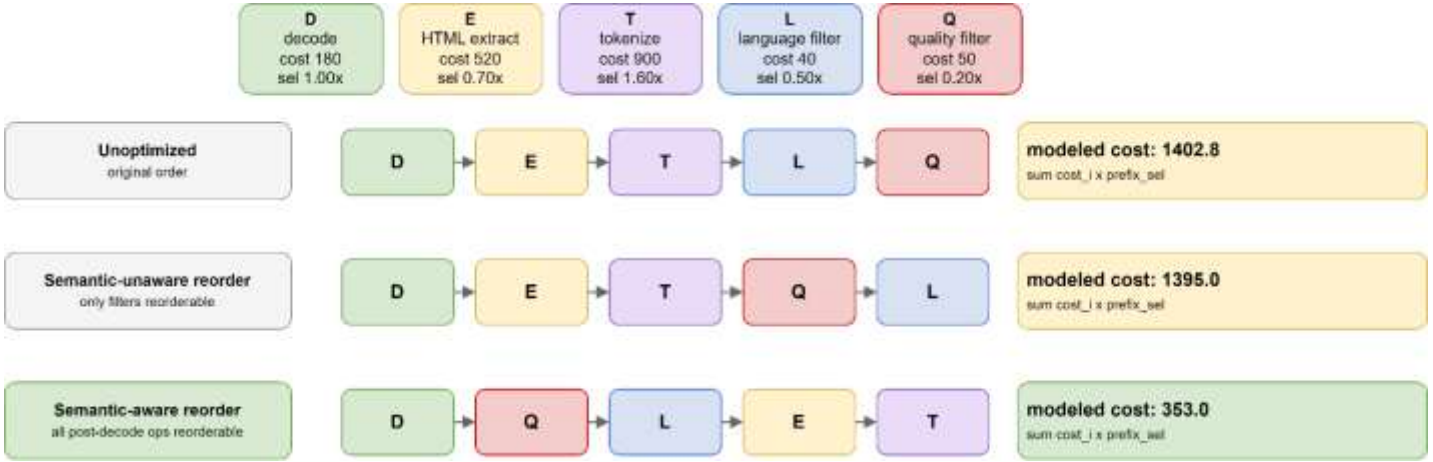


Optimization Opportunities in Input Pipelines

- Prior input systems improve throughput with techniques such as parallelism, prefetching, caching, offloading, fusion, and operator reordering.
  - Operator reordering changes the logical order in which transformations are applied.
  - Caching materializes the output of a deterministic prefix.
  - Offloading assigns an operator or operator block to an execution backend.
  - Fusion combines adjacent compatible operators into one physical unit.

Semantic-Aware Operator Reordering

- Semantic-aware reordering uses dependency and randomness annotations to identify which operator orders are fixed and which can be safely changed.
- This enables the optimizer to explore profitable reorderings without relying on relational algebra assumptions or risking invalid transformations.



Cost-Based Input Pipeline Optimization

- A cost-based optimizer provides a common basis for comparing these heterogeneous decisions.
- It first profiles the pipeline to estimate operator cost and selectivity, then uses these measurements to assign a modeled cost to each legal physical pipeline.
- Cedar is the representative cost-based optimizer for this setting.

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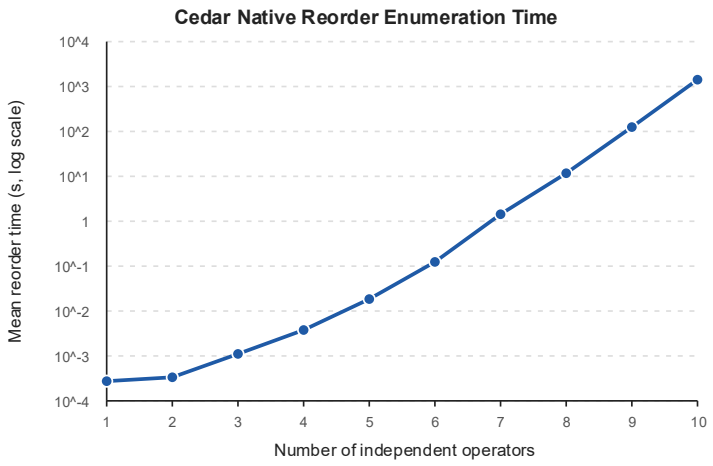
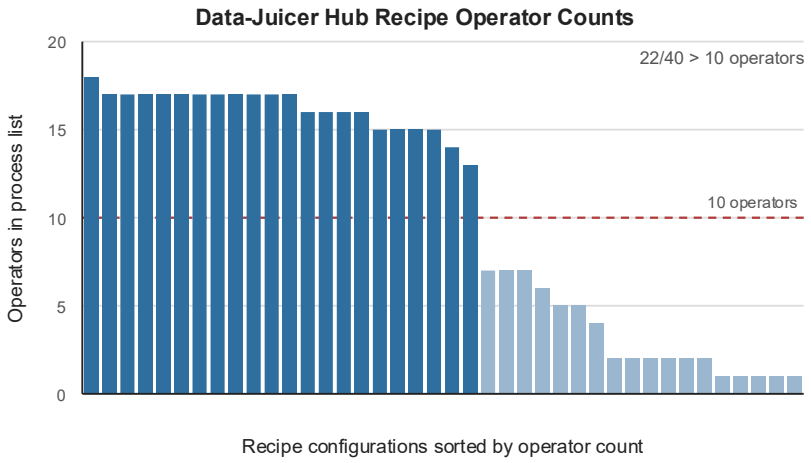
Semantic-Aware Operator Reordering Overhead

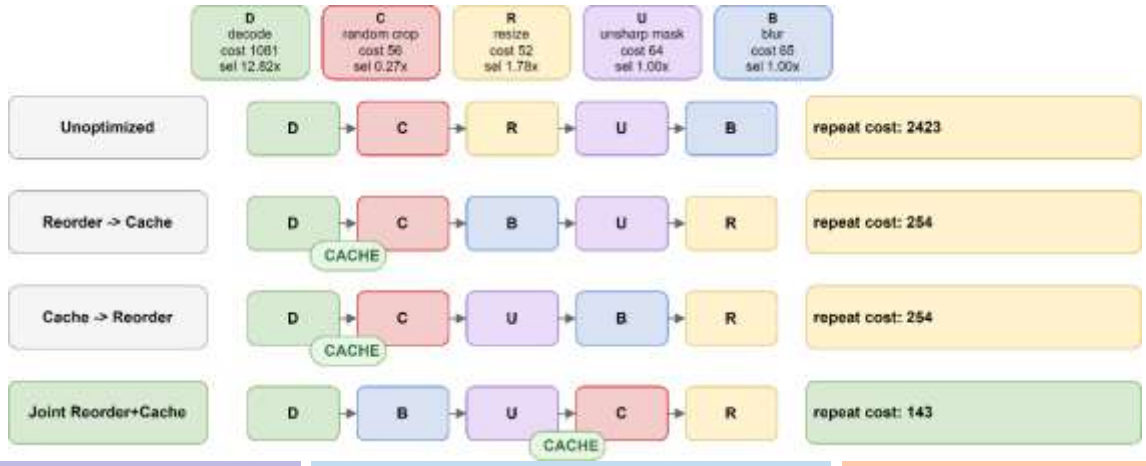
- Semantic-aware reordering search over all legal operator orders. Even with dependency constraints, the number of possible orders can grow factorially.
- This becomes expensive as input pipelines contain more reorderable operators.





Co-Optimization in Input Pipelines

- Input pipeline optimizations are tightly coupled.
- Stage-wise optimization may commit to a locally good plan too early.
- Joint optimization keeps these choices in the same search space, allowing the optimizer to find globally better pipeline plans.



Co-Optimization in Input Pipelines

- 虽然联合优化很好，但由于语义感知重排本身的开销较大，即使面向比较小的pipeline，我们也无法将其与其他优化技术做联合优化，因为这会导致较高的优化开销。

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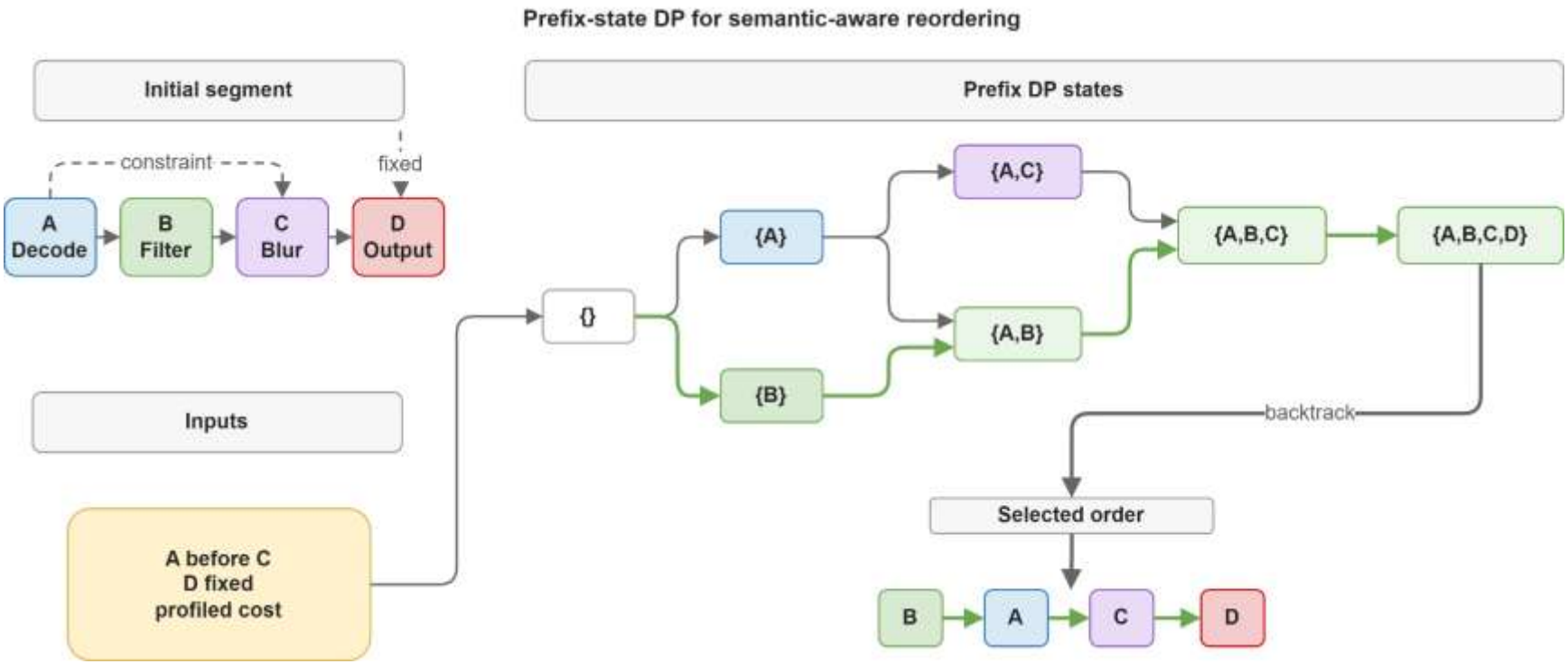
Opportunity: Prefix-State DP

- Many candidate orders share the same intermediate prefixes.
- Instead of enumerating every complete order, PICO represents search by the set of operators already placed.
- Each prefix state is evaluated once and reused by later extensions.

Prefix-State Dynamic Program

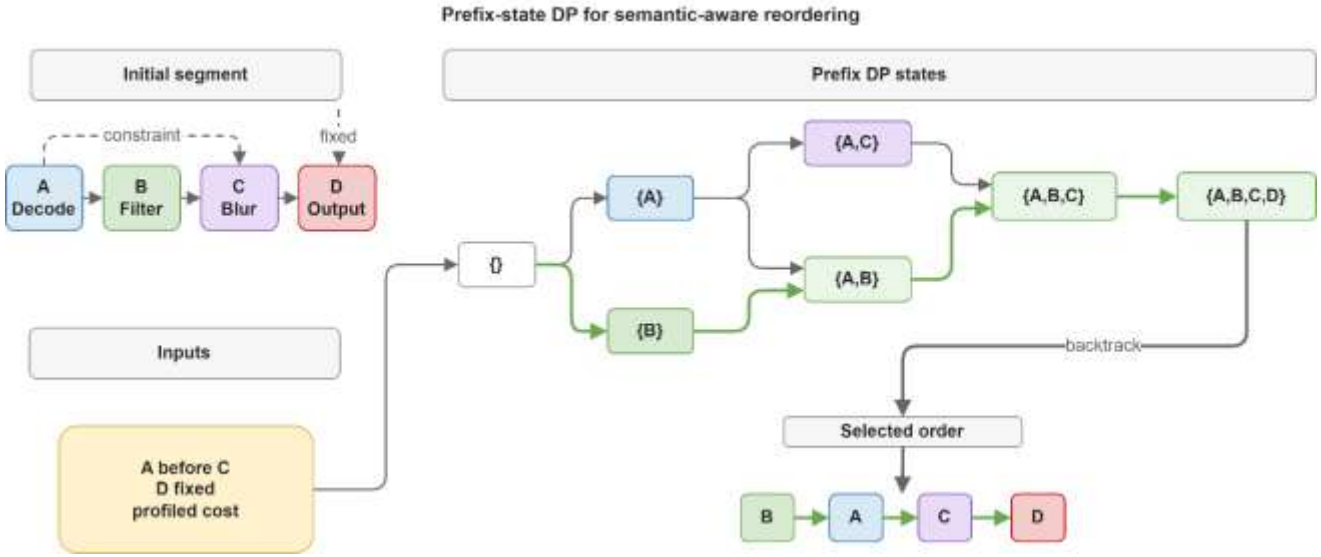


Reordering Problem Setup

- PICO optimizes reorderable linear segments in an input pipeline.
- Fixed boundaries and semantic dependencies define the legal search space.
- Profiling provides operator costs, input sizes, and data-volume ratios.
- The goal is to find the lowest-cost legal order under the profiled cost model.

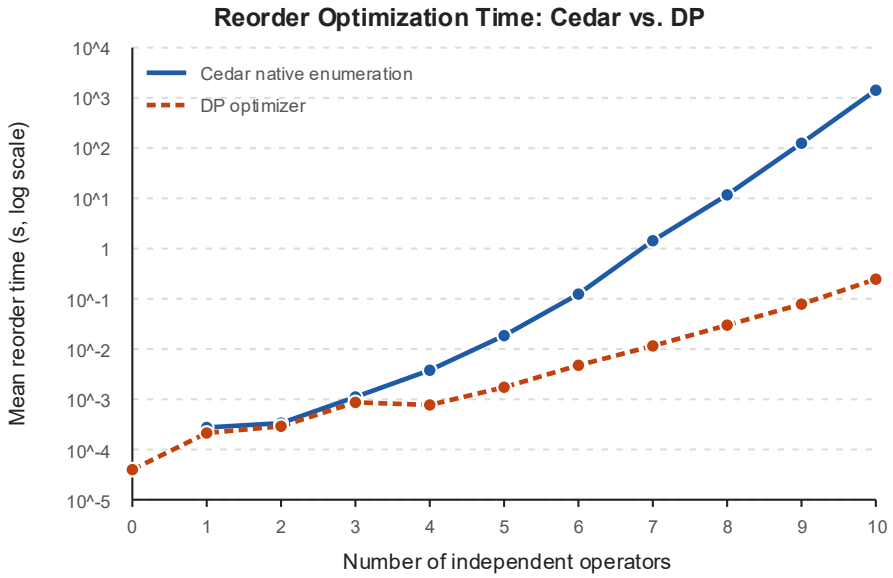
Prefix-State Dynamic Program

- Many candidate orders share the same set of already placed operators. Under the cost model, this set determines future data volume and legal next choices.
- PICO keeps only the cheapest prefix for each operator set. Backpointers reconstruct the final minimum-cost legal order.



Properties

- PICO searches subset states instead of complete permutations.Reordering takes O(n * 2^n) time, avoiding factorial enumeration.
- The optimality guarantee is scoped to the given semantic constraints and profiled cost model.



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Integrated DP View

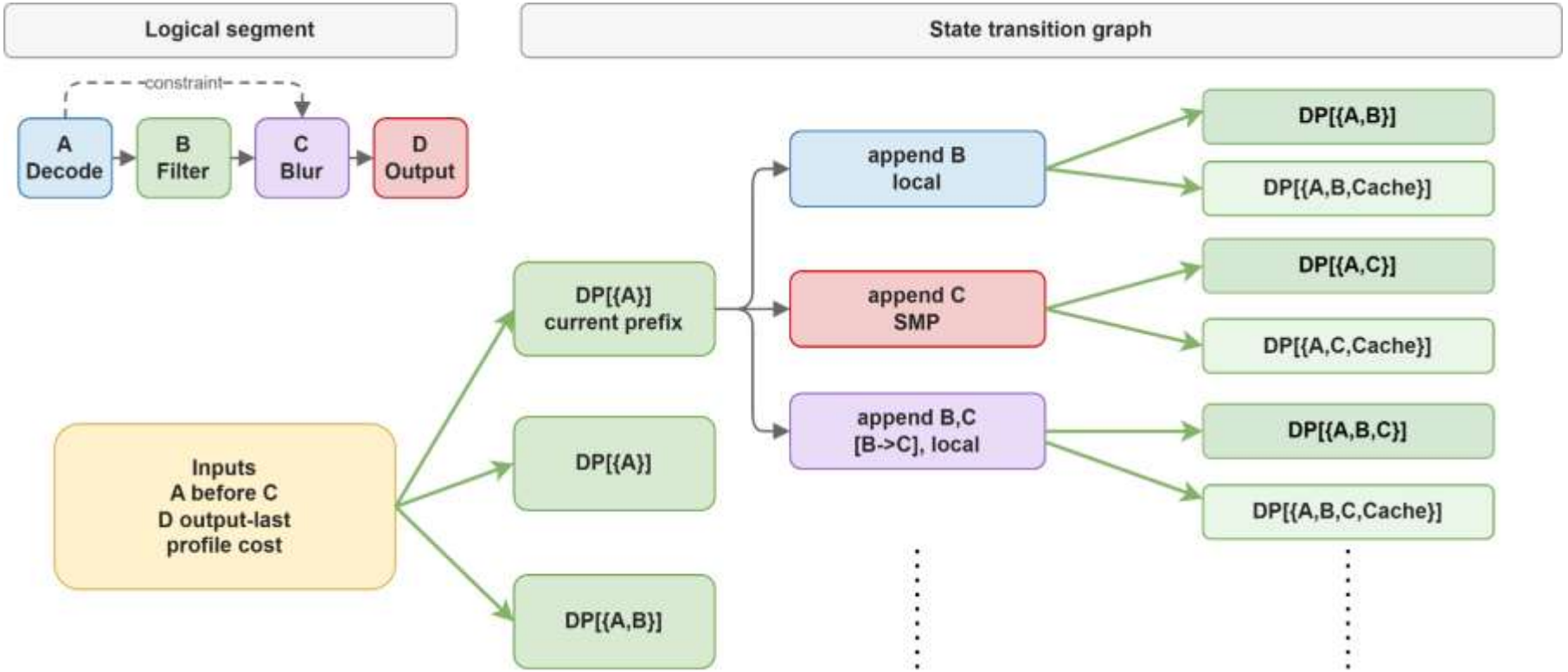
- PICO extends prefix-state DP from logical reordering to physical pipeline optimization.
- Each transition represents one physical planning decision.
- A transition may append a singleton operator or fused block, select a physical variant, and update cache state.
- Reordering, caching, offloading, and fusion are compared in the same DP.

Transition Candidates

- Before the outer DP, PICO precomputes feasible block candidates.
- For each variant, an inner DP finds the minimum-cost internal order for every block.
- PICO then compares feasible variants for the same block and keeps the lowest-cost implementation.
- The outer DP consumes each candidate as a fixed block with a known order, variant, and cost.

Transition Candidates

- The outer DP consumes each candidate as a fixed block with a known order, variant, and cost.



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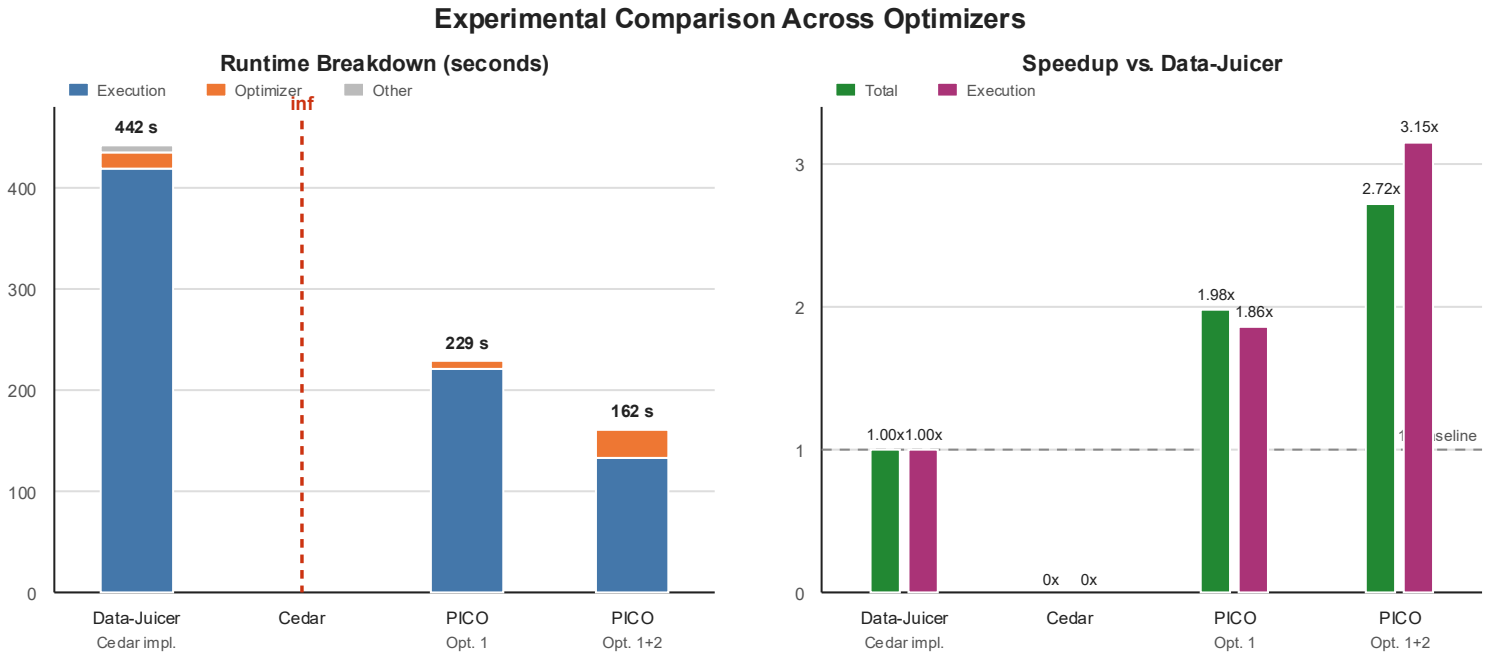
Experiment

- Bloom_Oscar workload with Wikitext03 dataset.



Thanks!